

SAKSHAM MRIG

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EDUCATION

Indraprastha Institute of Information Technology, Delhi
BTech in Computer Science and Applied Mathematics | CGPA – 8.68/10

New Delhi, India
2019 – 2023

WORK EXPERIENCE

Atlassian | *Software company that develops collaboration tools ([Jira](#) & [Confluence](#)) for developers and PMs*
Associate Data Scientist

Bengaluru, India
Jun 2023 – Present

- Optimized allocation of cloud-infrastructure for enterprise customers to enhance user experience
 - Computed user-behaviour, instance shape and cloud-infrastructure/geo-location segments for current cloud customers; performed a cohort analysis study using Amplitude to understand drivers of performance issues faced by end-users, identified by Shapley values
 - Collaborated with product managers to optimize allocation of appropriate cloud-infrastructure on a per-customer basis, reducing cloud residency cost for 200+ enterprise customers by ~ 8% and decreasing their average page response time by 15%
- Built a cloud trial-to-sales propensity model for improved audience targeting
 - Calculated historical “Cloud-Trial” adoption likelihood, conversion rates from past sales data and user activity statistics, and further passed these features into a Bayesian Model to determine the likelihood of adoption for prospective customers
 - Assisted the Demand Generation Team to identify customers with a higher potential upside (+43% conversion rate); highlighted under-utilisation of trials in specific cohorts and unexplored customer segments to drive migration sales
- Forecasted data storage and operational costs to strategize the pricing and packaging of a new Backup/Restore Product
 - Developed an Multiple Linear Regression (MLR) model using past product adoption and monthly usage data of 300K+ enterprise and SMB customers to predict future operational costs through different phases of a new Backup/Restore Product
 - Utilized these projections, along with competitor analysis, to build an optimized monthly usage-based pricing and packaging strategy; saved more than \$1.5k /customer/month in potentially unused AWS storage costs for Atlassian
- Established a fully automated source-control framework for all data science ETLsW developed by the team
 - Facilitated automatic syncing between the code repository and the data warehouse employing Bitbucket Pipelines, AWS and Apache Hive API; migrated 50+ legacy ETLs to the revamped architecture, making subsequent collaborations efficient
 - Built a Data-Quality Recommendation system, using Anthropic’s Claude 3.5, fine-tuned on internal SQL tables to suggest and implement fundamental data quality checks and save 10+Hr/Week of developer time
- Spearheaded design and execution of A/B tests for Cloud Migration Assistants’ performance and user-experience updates; utilized statistical significance tests to evaluate effectiveness of new features in improving performance metrics like Page Response Time
- Instrumented a novel metric, via a logical combination of product-operational events and sales/revenue data, that measures the dollar-impact of Cloud Migrations to help leadership better evaluate the performance of our migration strategy; used by the finance team in our Quarterly Earnings Release to showcase the impact of company’s cloud-migration efforts
- Formulated 4+ key dashboards in support of Level (1/2/3) OKRs using Python, SQL and Databricks/Tableau for senior management to showcase cloud migration efforts across sales and finance teams, and channel partners

Data Science Intern
2022

May – Aug

- Recommended the best cloud migration strategy for customers, by mapping business logic and guardrails used by Enterprise Partners into a Decision Tree, including factors such as Data complexity, Tentative downtime and other Technical specs
- Suggested the best options tailored for smaller customers by using past migration data from SMB Customers to train a Random Forest Classifier; achieved an Average Precision score of 0.70 and helped save hours of effort and time for Enterprise Partners
- Engineered a dashboard employing a pipeline using PySpark, Databricks and Plotly that gathers post migration-to-cloud survey responses, to identify and rank topics, using the Empath Topic Modelling algorithm, to gauge key drivers of customer satisfaction
- Synthesized data from the migration survey responses into 20+ insights, such as ranking partners based on feedback from customers, time spent during the various phases of the migration, etc.; shared reports with management on how to generate firm-wide impact

Google Summer of Code | *Google-funded global program for students to contribute to Open Source Software*
Software Developer Intern | [Github](#)

Remote
Apr – Aug 2020

- Refactored the legacy monolithic codebase to follow an MVC (Model-View-Controller) architecture by implementing an API layer, using Javascript, to make the method calls asynchronous and the core interpreter faster; streamlined codebase for future development
- Added 10+ new blocks in the visual programming interface like “Virtual Piano”, an on-screen piano and “Oscilloscope” block, used by 2K+ instructors and learners as a free guitar tuner
- Resolved 71 github-bugs and delivered feature requests like an online project sharing platform for students built on a MongoDB database, using NLP, to improve in-app block search by 24%; launched an offline mobile version for 100+ students in Peru

PUBLICATIONS

Leveraging Intra and Inter Modality Relationship for Multimodal Fake News Detection

Apr 2022

WWW' 22: The ACM Web Conference | Lyon, France | [Github](#)

- Built a Fake News Detection framework trained on two publicly available multimodal (Images and Text) datasets - “Verifying Multimedia Use - Challenge” and “Weibo Fake News”, exploiting a blend of intra and inter modality relationships
- Presented a novel architecture that identifies and suppresses information from weaker modalities and extracts relevant information from strong modality on a per-sample basis; outperformed existing methodologies by **3.05%** (accuracy) and **4.53%** (F1-score)
- Captured the intra-modality relationship in images via a bottom-up attention approach using an R-CNN framework, to identify only the salient patches from images
- Extracted the phrase level information from the text using their Unigram/Bigram/Trigram representations, finally passing the text through a self-attention module to obtain intermediate embeddings

ACHIEVEMENTS

Finalist, Mathematical Contest in Modelling (MCM/ICM)

Feb 2023

- Represented IIITD in the annual modelling competition organized by [COMAP](#): Consortium for Mathematics and its Applications; ranked **3rd in India** and within the **top 3% globally** among 25,000 teams
- Developed a framework that can estimate the average number of tries required to complete the NY Times WORDLE puzzle
- Combined a word difficulty predictor, built using XGBoost classifier trained on historical words and their difficulty, with a Dirichlet Regression Model that can model the number of tries required to successfully complete the puzzle; achieved a brier score of <0.1
- Performed a university-wide WORDLE survey to model 36 different strategies used by players with varying levels of expertise into an extensive Hierarchical Markov Chain, which can get the correct answer within 4 tries 7 out of 10 times
- Employed Interpretability frameworks like LIME and SHAP to understand what attributes of the word make it difficult for people to guess and presented the findings in the Solution Paper

1st Place, HackPSU | [Github](#)

Mar 2021

- Won first position among **515** international teams at HackPSU (organized by Penn State University); also won the Nittany Ai Alliance “Ai for Good” Challenge, winning a total sum of **\$1000**
- Built a browser-extension based personal productivity companion app that tracks browsing activity via the Chrome-extension V2 API and reminds users whenever they are going off-topic or procrastinating, while also reminding users at set intervals to take short breaks
- Used the Microsoft Azure Facial Emotion model to track user’s facial expressions while browsing; generated a daily analysis report of their browsing activity alongside the emotions captured from their face, helping user’s maintain mental well-being

ACADEMIC RESEARCH AND PROJECTS

Safe Exploration in Deep Reinforcement Learning

Aug - Dec 2022

- Designed and developed a RL Agent that follows a zero constraint violation policy during the exploration/learning phase in environments where certain critical constraints can’t be violated, like in the case of autonomous vehicles and data center cooling systems, using a state-based action correction layer on top of Standard Deep RL Algorithms like DDPG and TD3
- Showcased the effectiveness of our Agent on Ball-3d and Spaceship Openai gym environments via faster convergence, 10-15% higher cumulative reward than other alternatives, while following a zero-violation policy

Detecting Causes of Mental Health Issues in Code-Mixed Social Media Posts

Aug - Dec 2022

- Scraped and augmented a dataset of 12,000 Reddit posts from top mental health subreddits; translated this data into Hindi/English code-mixed using a pretrained multilingual hi-en version of mBART to test the Multilingual robustness of the model
- Identified underlying causes of mental illness by analysing the user’s social media posts and training a Sentence-BERT model on the code-mixed data along with the CAMS dataset; gained **+5%** in accuracy over alternative benchmark models

PythonTutor ++ | [Github](#) | [Online Tool URL](#)

Aug - Dec 2022

- Developed an advanced version of Pythontutor.com, a program visualization tool widely-used in CS education, with additional functionality to analyze ByteCode execution and Evaluation Stack operations at the heart of Python runtime
- Extracted the Internal Symbols, AST and Evaluation Stack from CPython’s runtime at each step of ByteCode execution and displayed it in an interactive NodeJs WebApp for easy visual learning and debugging during the execution of Python Bytecode
- Deployed the website for 200+ students studying “Intro to Programming” and “Compilers” courses at IIIT Delhi

POSITIONS OF RESPONSIBILITY

President: Byld (Software development club @ IIIT Delhi)

Aug 2022 - Feb 2023

- Led a team of 7 Students to organize a 2-Day technical hackathon with talks from industry leaders on exciting new fields for a cohort of 200+ student attendees and 70 participating teams
- Redesigned the official [IIITD library portal](#) using Svelte and headless-Drupal (CMS), reducing the page load time by 32%; devised a framework for website managers to create responsive and interactive webpages using simple markdown and yaml files
- Facilitated the pairing and outreach of a mentor-mentee cohort of 30+ freshman/sophomore students interested in upskilling technically; resulted in 3-Google Summer of Code, 2-Outreachy and 1-FOSSAsia internship acceptances

TECHNICAL SKILLS

Programming: R, Python, C/C++

Data Science: SQL, Tableau, MATLAB, Apache Spark, Pandas, NumPy, Seaborn

Machine Learning Frameworks: Scikit-Learn, PyTorch, Keras

ADDITIONAL INFORMATION

Community Engagement: Participated in trash clean-up activities in community parks, Bengaluru (Goodera, May 2024); volunteered in local government schools to teach elementary mathematics and serve mid-day meals, Bengaluru (KV DRDO School, July 2023);

Online Courses: Neural Networks and Deep Learning (Coursera, 2020); Haskell MOOC (University of Helsinki, 2023)

Interests: Officially ranked 29th (by the [SREI](#)) in U17 Junior Squash; Led the IIITD Squash team to inter-college victory in 2023; enjoy playing Keyboard (Grade 3, Trinity College London); completed the “Developing Your Musicianship” Specialization (Berklee 2020)